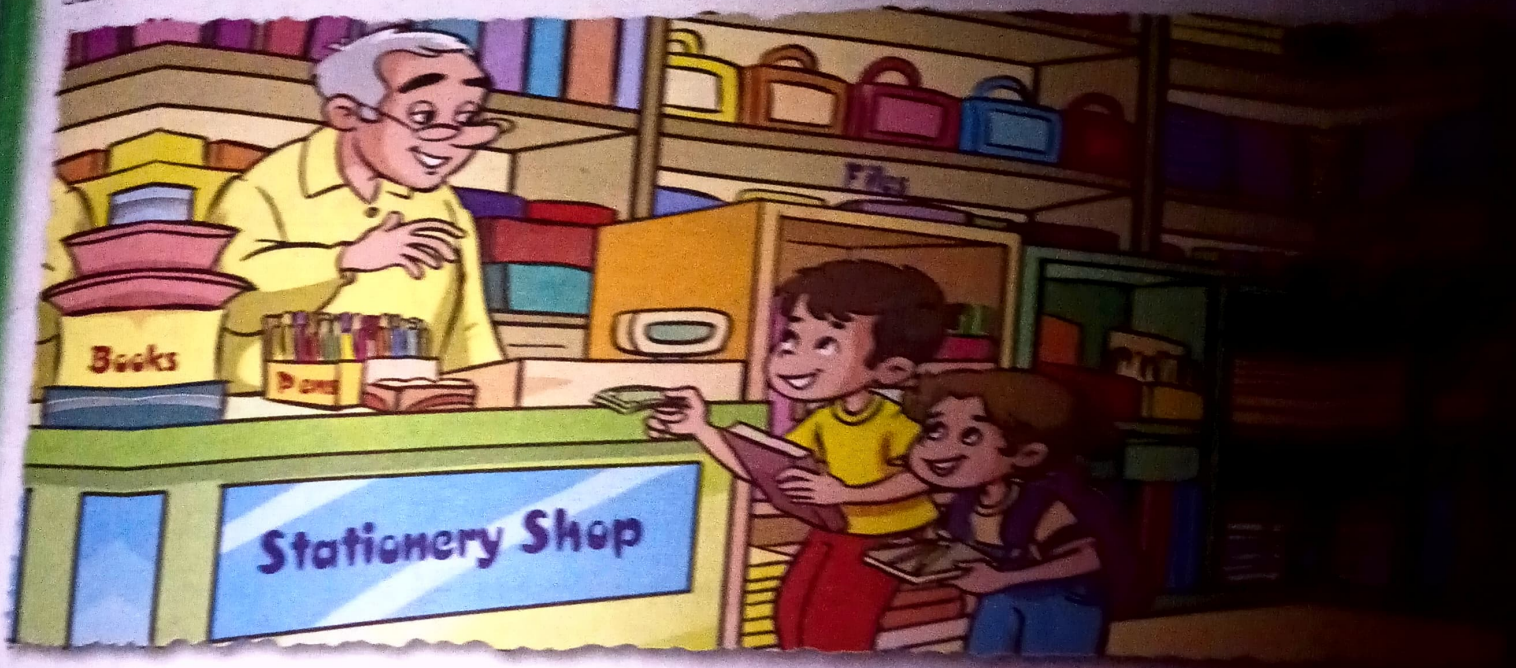
Embark On

Seema wants to buy three pens, each costing ₹ 5. Her brother Sachin wants to buy similar pens. They decided to go to a wholesale shop to make their purchase.

At the shop, the shopkeeper tells them that a packet of 5 pens costs ₹ 22. Now, Seema and Sachin need to figure out the cost of each pen to ensure they are getting a good deal.



Seema knows that each pen costs ₹ 5 when bought individually.

However, the shopkeeper offers a packet of 5 pens for ₹ 22.

To find out the cost of each pen when bought in a packet, we divide the total cost of the packet by the number of pens in it.

Mathematically, this is represented as:

cost of each pen = ₹ $\frac{22}{5}$

Dividing ₹ 22 by 5 gives us: $\frac{22}{5}$ = ₹ 4.40

Therefore, each pen in the packet costs ₹ 4.40.

The division resulted in a number that isn't a whole number, but a decimal. Specifically ₹ 4.40.

Here, ₹ 4.40 is a rational number because it can be expressed as a fraction where the numerator (22) and the denominator (5) are integers.

1

CHAPTER

Rational Numbers

Embark On

NCF: Direct Perception
NEP: Analytical Thinking

Seema wants to buy three pens, each costing ₹ 5. Her brother Sachin wants to buy two similar pens. They decided to go to a wholesale shop to make their purchase.

At the shop, the shopkeeper tells them that a packet of 5 pens costs ₹ 22. Now, Seema and Sachin need to figure out the cost of each pen to ensure they are getting a good deal.



Seema knows that each pen costs ₹ 5 when bought individually.

However, the shopkeeper offers a packet of 5 pens for ₹ 22.

To find out the cost of each pen when bought in a packet, we divide the total cost of the packet by the number of pens in it.

Mathematically, this is represented as:

cost of each pen = $₹ \frac{22}{5}$

Dividing ₹ 22 by 5 gives us: $\frac{22}{5} = ₹ 4.40$

Therefore, each pen in the packet costs ₹ 4.40.

The division resulted in a number that isn't a whole number, but a fraction or decimal, specifically ₹ 4.40.

Here, ₹ 4.40 is a rational number because it can be expressed as a fraction where both the numerator (22) and the denominator (5) are integers.

Rational Numbers

A rational number is a number that can be expressed as a quotient or fraction $\frac{p}{q}$ of two integers, whe
p (the numerator) is an integer and q (the denominator) is a non-zero integer.

Rational numbers are numbers that can be written in the form of $\frac{p}{q}$ where p and q are integers an
$q \neq 0$.

- Any integer is a rational number, as it can be expressed as $\frac{a}{1}$, where a is an integer.
- Rational numbers include both positive and negative numbers, as well as zero.
- Rational numbers can be represented on the number line.
- Every fraction is a rational number.

Operations on Rational Numbers:

Addition: If $\frac{a}{b}$ and $\frac{c}{b}$ are two rational numbers with the same denominator b, then,

$$\frac{a}{b} + \frac{c}{b} = \frac{a+c}{b}.$$

For example, $\frac{5}{3} + \frac{4}{3} = \frac{5+4}{3} = \frac{9}{3} = 3.$

Now, if $\frac{a}{b}$ and $\frac{c}{d}$ are two rational numbers such that $b \neq d$, then $\frac{a}{b} + \frac{c}{d} = \frac{ad+bc}{bd}$.

For example, $\frac{-9}{7} + \frac{5}{8} = \frac{-9 \times 8 + 7 \times 5}{7 \times 8} = \frac{-72+35}{56} = \frac{-37}{56}$.

Additive Inverse: If $\frac{a}{b}$ and $\frac{c}{d}$ are two rational numbers such that, $\frac{a}{b} + \frac{c}{d} = 0$, then $\frac{a}{b}$ is called the additive inverse of $\frac{c}{d}$ and vice-versa. For exemple, $\frac{a}{b} + \frac{-a}{b} = 0$.

It follows that additive inverse of $\frac{3}{7}$ is $\frac{-3}{7}$ and additive inverse of $\frac{-3}{7}$ is $\frac{3}{7}$.

Subtraction: If $\frac{a}{b}$ and $\frac{c}{d}$ are two rational numbers, then subtracting $\frac{c}{d}$ form $\frac{a}{b}$ means adding additive inverse of $\frac{c}{d}$ and $\frac{a}{b}$.

The subtraction of $\frac{c}{d}$ from $\frac{a}{b}$ is written as $\frac{a}{b} - \frac{c}{d}$.

For example, $\frac{-4}{9}$ from $\frac{-5}{7} \Rightarrow \frac{-5}{7} - \left(\frac{-4}{9}\right) = \frac{-5}{7} + \frac{4}{9} = \frac{-45+28}{63} = \frac{-17}{63}$

Multiplication: Product of two rational numbers = $\frac{\text{Product of their numerators}}{\text{Product of their denominators}}$

If $\frac{a}{b}$ and $\frac{c}{d}$ are two rational numbers then, $\frac{a}{b} \times \frac{c}{d} = \frac{ac}{bd}$.

Reciprocal of a Rational Number:

If $\frac{a}{b}$ and $\frac{c}{d}$ are two rational numbers such that $\frac{a}{b} \times \frac{c}{d} = 1$ then $\frac{a}{b}$ is called the reciprocal of $\frac{c}{d}$ an

Division: Division is the inverse of multiplication. If $\frac{a}{b}$ and $\frac{c}{d}$ are two rational numbers such that $\frac{c}{d} \neq 0$, then,

$$\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \left(\text{reciprocal of } \frac{c}{d}\right) = \frac{a}{b} \times \frac{d}{c}$$

For example: Divide $\frac{9}{11}$ by $\frac{3}{5} = \frac{9}{11} \div \frac{3}{5} = \frac{9}{11} \times \frac{5}{3} = \frac{9 \times 5}{11 \times 3} = \frac{15}{11}$.



Exercise - 1A

1. Add the following rational numbers:

(a) $\frac{5}{2}$ and $\frac{-11}{2}$ (b) $\frac{7}{5}$ and $\frac{13}{5}$ (c) $\frac{-7}{8}$ and $\frac{12}{-8}$ (d) $\frac{-11}{5}$ and $\frac{-9}{5}$

(e) $\frac{4}{-7}$ and $\frac{-5}{8}$ (f) $\frac{-13}{7}$ and $\frac{12}{-5}$ (g) $\frac{-15}{9}$ and $\frac{-17}{11}$ (h) $\frac{-9}{-33}$ and $\frac{5}{11}$

2. Subtract the following rational numbers:

(a) $\frac{-18}{5} - \frac{-8}{5}$ (b) $\frac{-19}{15} - \frac{-6}{30}$ (c) $\frac{-6}{11} - \frac{-7}{11}$ (d) $\frac{-1}{4} - \frac{-2}{8}$

(e) $\frac{-59}{5} - \frac{19}{5}$ (f) $\frac{-4}{7} - \frac{-8}{9}$ (g) $\frac{7}{24} - \frac{-13}{-26}$ (h) $\frac{7}{25} - \frac{-6}{15}$

3. Multiply the following:

(a) $\frac{4}{7}$ by $\frac{-2}{5}$ (b) $\frac{-3}{8}$ by $\frac{-12}{15}$ (c) $\frac{11}{-13}$ by $\frac{-39}{22}$ (d) $\frac{-5}{9}$ by $\frac{81}{35}$

(e) $\frac{-9}{25}$ by $\frac{-35}{27}$ (f) $\frac{-18}{25}$ by $\frac{40}{-36}$

4. Divide the following:

(a) $\frac{16}{7}$ by $\frac{-8}{14}$ (b) $\frac{-7}{8}$ by -21 (c) $\frac{3}{8}$ by $\frac{-4}{5}$ (d) $\frac{-1}{15}$ by $\frac{8}{3}$

(e) $\frac{-3}{26}$ by $\frac{9}{78}$ (f) $\frac{-22}{26}$ by $\frac{-33}{39}$

5. Simplify:

(a) $\left(\frac{3}{2} \times \frac{-7}{4} \times \frac{8}{9}\right) - \left(\frac{-15}{2} \times \frac{3}{7} \times \frac{8}{14}\right)$ (b) $\left(\frac{7}{3} \times \frac{9}{11}\right) + \left(\frac{4}{3} \times \frac{7}{22}\right) - \left(\frac{3}{-11} \times \frac{-7}{9}\right)$

Since, addition of rational numbers produces rational numbers, we say that the collection of ratio numbers is closed under addition. This property is called the closure property for addition of ratio numbers.

For example, $\frac{3}{4}+\frac{7}{3}=\frac{9+28}{12}=\frac{37}{12}$.

Commutative Property: If $\frac{a}{b}$ and $\frac{c}{d}$ are two rational numbers, then $\frac{a}{b}+\frac{c}{d}=\frac{c}{d}+\frac{a}{b}$.

We say that addition of rational numbers is commutative. This property is called commutative prope for addition of rational numbers.



For example, $\frac{3}{8}+\frac{5}{7}=\frac{5}{7}+\frac{3}{8}$

$\Rightarrow \frac{21+40}{56}=\frac{40+21}{56}$ $\Rightarrow \frac{61}{56}=\frac{61}{56}$

$\therefore \frac{3}{8}+\frac{5}{7}=\frac{5}{7}+\frac{3}{8}$.

Hence, commutative property for addition of rational numbers is verified.

Associative Property : If $\frac{a}{b}, \frac{c}{d}$ and $\frac{e}{f}$ are three rational numbers,

then $\left(\frac{a}{b}+\frac{c}{d}\right)+\frac{e}{f}=\frac{a}{b}+\left(\frac{c}{d}+\frac{e}{f}\right)$

We say that rational numbers are associative under addition. This property is known as associati property for addition of rational numbers.

For example, Verify: $\frac{-2}{5}+\left(\frac{3}{4}+\frac{-7}{8}\right)=\left(\frac{-2}{5}+\frac{3}{4}\right)+\frac{-7}{8}$



$\Rightarrow \frac{-2}{5}+\left(\frac{6+(-7)}{8}\right)=\left(\frac{-8+15}{20}\right)+\frac{-7}{8}$

$\Rightarrow \frac{-2}{5}+\frac{-1}{8}=\frac{7}{20}+\frac{-7}{8}$

$\Rightarrow \frac{-16-5}{40}=\frac{14-35}{40}$

$\Rightarrow \frac{-21}{40}=\frac{-21}{40}$

Hence, $\frac{-2}{5}+\left(\frac{3}{4}+\frac{-7}{8}\right)=\left(\frac{-2}{5}+\frac{3}{4}\right)+\frac{-7}{8}$.

Additive Property of Zero (0):

If $\frac{a}{b}$ is any rational number, then $0+\frac{a}{b}=\frac{a}{b}+0=\frac{a}{b}$.

If we add 0 to any rational number or any rational number is added to 0, the rational numb remains the same. Therefore '0' is called the additive identity of rational numbers.

For example, $0+\frac{2}{5}=\frac{2}{5}$

$\Rightarrow \quad \frac{2}{5} = \frac{2}{5} + 0 \qquad \Rightarrow \frac{2}{5} = \frac{2}{5}$

Hence, $0 + \frac{2}{5} = \frac{2}{5} + 0 = \frac{2}{5}$.

Properties of Subtraction of Rational Numbers

Closure Property:

The result of subtraction of one rational number from another is again a rational number.

If $\frac{a}{b}$ is subtracted from $\frac{c}{d}$, then $\frac{c}{d} - \frac{a}{b}$ is again a rational number. We say that the collection of rational numbers is closed under subtraction. This property is known as closure property for subtraction of rational numbers.

For example, $\frac{-3}{5} - \left(\frac{-7}{6}\right) = \frac{-3}{5} + \frac{7}{6} = \frac{-18 + 35}{30} = \frac{17}{30}$, which is also a rational number.

Property of Zero (0):

If 0 is subtracted from a rational number, the result is the rational number itself.

If 0 is subtracted from a rational number $\frac{a}{b}$, then the result is $\frac{a}{b}$, i.e., $\frac{a}{b} - 0 = \frac{a}{b}$.



Exercise - 1B

Fill in the blanks using commutative property for addition of rational numbers:

(a) $\frac{-3}{7} + \frac{4}{9} =$ _____

(b) $\frac{2}{3} + \left(\frac{-5}{6}\right) =$ _____

(c) $\left(\frac{-11}{29}\right) + \left(\frac{-5}{31}\right) =$ _____

(d) $\frac{-7}{13} + \frac{11}{23} =$ _____

Fill in the blanks using associative property for addition of rational nunmbers:

(a) $\left(\frac{1}{11} + \frac{2}{13}\right) + \frac{7}{6} =$ _____

(b) $\frac{17}{21} + \left[\left(\frac{-5}{13}\right) + \frac{9}{16}\right] =$ _____

(c) $\left(\frac{-31}{41}\right) + \left(\frac{9}{14} + \frac{8}{15}\right) =$ _____

(d) $\left(\frac{2}{7} + \frac{3}{8}\right) + \frac{-9}{14} =$ _____

State the property used in each of the following:

(a) $\frac{-2}{5} + \frac{3}{7} = \frac{3}{7} + \frac{-2}{5}$

(b) $\frac{2}{5} + \left(\frac{9}{2} + \frac{-3}{8}\right) = \left(\frac{2}{5} + \frac{9}{2}\right) + \frac{-3}{8}$

(c) $\frac{-3}{7} + \left(\frac{-5}{8} + \frac{9}{4}\right) = \left(\frac{-3}{7} + \frac{-5}{8}\right) + \frac{9}{4}$

(d) $\frac{3}{10} + \left(\frac{-11}{15} + \frac{9}{-7}\right) = \left(\frac{3}{10} + \frac{-11}{15}\right) + \frac{9}{-7}$

If $a = \frac{8}{9}$ and $b = \frac{-3}{8}$, then verify that $a + b = b + c$.

Properties of Multiplication of Rational Numbers

Closure Property:

The product of two rational numbers is a rational number. Since multiplication of rational numb produce rational numbers, we say that the collection of rational numbers is closed under multiplicati This property is known as closure property for multiplication of rational numbers.

For example, $\frac{3}{4} \times \frac{2}{7} = \frac{6}{28} = \frac{3}{14}$, which is a rational number.

Commutative Property:

If $\frac{a}{b}$ and $\frac{c}{d}$ are two rational numbers, then $\frac{a}{b} \times \frac{c}{d} = \frac{c}{d} \times \frac{a}{b}$. We say that rational numbers are commutati under multiplication. This property is known as commutative property for multiplication of ration numbers.

For example, $\frac{3}{4} \times \frac{8}{26} = \frac{3 \times 8}{4 \times 26} = \frac{24}{104} = \frac{3}{13}$. Also $\frac{8}{26} \times \frac{3}{4} = \frac{8 \times 3}{26 \times 4} = \frac{24}{104} = \frac{3}{13}$.

Associative Property:

It $\frac{a}{b}$, $\frac{c}{d}$ and $\frac{e}{f}$ are three rational numbers, then, $\left(\frac{a}{b} \times \frac{c}{d}\right) \times \frac{e}{f} = \frac{a}{b} \times \left(\frac{c}{d} \times \frac{e}{f}\right)$

We say that rational numbers are associative under multiplication.

This property is known as associative property for multiplication of rational numbers.

For example, $\left(\frac{-2}{5} \times \frac{3}{7}\right) \times \frac{3}{8} = \left(\frac{-6}{35}\right) \times \frac{3}{8} = \frac{-18}{280} = \frac{-9}{140}$.

Also, $\frac{-2}{5} \times \left(\frac{3}{7} \times \frac{3}{8}\right) = \frac{-2}{5} \times \frac{9}{56} = \frac{-18}{280} = \frac{-9}{140}$.



Multiplicative Identity:

$\frac{a}{b}$ is a rational number, then $1 \times \frac{a}{b} = \frac{a}{b}$.

, 1 is called multiplicative identity of rational numbers. This is also called property of 1.

r example, $1 \times \frac{3}{8} = \frac{3}{8}$ and $\frac{3}{8} \times 1 = \frac{3 \times 1}{8} = \frac{3}{8}$.

operty of Zero (0):

is a rational number, then $0 \times \frac{a}{b} = 0 = \frac{a}{b} \times 0$.

example, $\frac{3}{11} \times 0 = \frac{3 \times 0}{11} = \frac{0}{11} = 0$.

ributive Property of Multiplication Over Addition or Subtraction:

$\frac{c}{l}$ and $\frac{e}{f}$ and are three rational numbers, then

$\frac{a}{b} \times \left(\frac{c}{d} + \frac{e}{f}\right) = \left(\frac{a}{b} \times \frac{c}{d}\right) + \left(\frac{a}{b} \times \frac{e}{f}\right) = \left(\frac{a \times c}{b \times d}\right) + \left(\frac{a \times e}{b \times f}\right)$

And $\frac{a}{b}\times\left(\frac{c}{d}-\frac{e}{f}\right)=\left(\frac{a}{b}\times\frac{c}{d}\right)-\left(\frac{a}{b}\times\frac{e}{f}\right)=\left(\frac{a\times c}{b\times d}\right)-\left(\frac{a\times e}{b\times f}\right)$

This property is known as distributive property of multiplication over addition or subtraction.

For example, Verify: $\frac{2}{5}\times\left(\frac{-3}{7}+\frac{6}{11}\right)=\left(\frac{2}{5}\times\frac{-3}{7}\right)+\left(\frac{2}{5}\times\frac{6}{11}\right)$



L.H.S $=\frac{2}{5}\times\left(\frac{-3}{7}+\frac{6}{11}\right)=\frac{2}{5}\times\left(\frac{-33+42}{77}\right)=\frac{2}{5}\times\frac{9}{77}=\frac{18}{385}$.

R.H.S $=\left(\frac{2}{5}\times\frac{-3}{7}\right)+\left(\frac{2}{5}\times\frac{6}{11}\right)=\frac{-6}{35}+\frac{12}{55}=\frac{-66+84}{385}=\frac{18}{385}$.

$\therefore$ L.H.S = R.H.S.

$\therefore$ $\frac{2}{5}\times\left(\frac{-3}{7}+\frac{6}{11}\right)=\left(\frac{2}{5}\times\frac{-3}{7}\right)+\left(\frac{2}{5}\times\frac{6}{11}\right)$

Properties of Division of Rational Numbers

(a) If $\frac{a}{b}$ and $\frac{c}{d}$ are two rational numbers, then $\frac{a}{b}\div\frac{c}{d}$ is always a rational number, $\frac{c}{d}\neq 0$. (Division by 0 is not defined)

(b) If $\frac{a}{b}$ is a rational number and $\frac{a}{b}\neq 0$, then $\frac{a}{b}\div\frac{a}{b}=1$.

$\frac{a}{b}\div\left(\frac{-a}{b}\right)=-1$ and $\left(\frac{-a}{b}\right)\div\frac{a}{b}=-1$.

(c) If $\frac{a}{b}$ is a rational number, then $\frac{a}{b}\div 1=\frac{a}{b}$, and $\frac{a}{b}\div(-1)=\frac{-a}{b}$.





Exercise - 1C

Find the product and verify the commutative property for multiplication of rational numbers:

(a) $\frac{1}{11}\times\frac{6}{7}$ (b) $\frac{3}{5}\times\left(\frac{-7}{8}\right)$ (c) $\left(\frac{-13}{19}\right)\times\frac{7}{8}$ (d) $\frac{-5}{9}\times\left(\frac{-11}{32}\right)$

Find the product and verify the associative property for multiplication of rational numbers:

(a) $\left(\frac{7}{20}\times\frac{5}{21}\right)\times\frac{1}{3}$ (b) $\frac{2}{7}\times\left(\frac{-7}{3}\times\frac{6}{-11}\right)$ (c) $\left(\frac{-7}{15}\times\frac{-2}{9}\right)\times\left(\frac{-3}{7}\right)$ (d) $\frac{8}{9}\times\left(\frac{-8}{5}\times\frac{6}{7}\right)$

Simplify the following and verify the distributive property of multiplication over addition:

(a) $\frac{5}{4}\times\left(\frac{-6}{7}+\frac{2}{5}\right)$ (b) $3\times\left[\frac{1}{3}+\left(\frac{-5}{11}\right)\right]$ (c) $0\times\left(\frac{0}{1}\times\frac{2}{5}\right)$

Write the name of the property used in each of the following:

(a) $\frac{2}{7}\times\frac{13}{11}=\frac{13}{11}\times\frac{2}{7}$

(b) $\frac{-5}{7}\times\left(\frac{6}{11}\times\frac{7}{13}\right)=\left(\frac{-7}{11}\times\frac{5}{8}\right)\times\frac{5}{13}$

(c) $\frac{1}{7}\times\frac{3}{5}=\frac{3}{35}$ is a rational number

(d) $\frac{7}{11}\times\left(\frac{1}{6}+\frac{2}{13}\right)=\frac{7}{11}\times\frac{1}{6}+\frac{7}{11}\times\frac{2}{13}$

(e) $\frac{-9}{8}\times 0=0=0\times\frac{-9}{8}$

(f) $1\times\frac{7}{9}=\frac{7}{9}$

5. Fill in the blanks:

(a) $\left(\frac{-9}{16}\right) \times \left(\frac{11}{7}\right) = \left(\frac{11}{7}\right) \times (____)$

(b) $(____) \times \left(\frac{-7}{9}\right) = \left(\frac{-7}{9}\right) \times \left(\frac{-5}{8}\right)$

(c) $\left(\frac{-7}{13}\right) \times _____ = \left(\frac{-7}{13}\right)$

(d) $____ \times \frac{-19}{47} = 0$

(e) $\frac{-3}{4} \times \left[\frac{1}{3} + \left(\frac{-5}{6}\right)\right] = \left(\frac{-3}{4} \times ____\right) + \left(\frac{-3}{4} \times ____\right)$

(f) $\frac{-2}{5} \times \left(\frac{6}{7} \times \frac{-8}{9}\right) = \left(\frac{-2}{5} \times ____\right) \times \frac{-8}{9}$

(g) $\frac{3}{8} \div \frac{3}{8} = _____$

(h) $\frac{7}{13} \div _____ = -1$

(i) $\frac{14}{19} \div ____ = \frac{14}{19}$

(j) $____ \div \left(\frac{-13}{15}\right) = 1$

Prescribed Questions with Solutions

1. Name the property under multiplication used in each of the following:

(i) $\frac{-4}{5} \times 1 = 1 \times \frac{-4}{5} = -\frac{4}{5}$ (ii) $-\frac{13}{17} \times \frac{-2}{7} = \frac{-2}{7} \times -\frac{13}{17}$ (iii) $\frac{-19}{29} \times \frac{29}{-19} = 1$

Sol. (i) $\frac{-4}{5} \times 1 = 1 \times \frac{-4}{5} = -\frac{4}{5}$, 1 is the multiplicative identity.

(ii) Commutativity (iii) Multiplicative inverse

2. Tell what property allows you to compute $\frac{1}{3} \times \left(6 \times \frac{4}{3}\right)$ as $\left(\frac{1}{3} \times 6\right) \times \frac{4}{3}$.

Sol. It is of the form, $a \times (b \times c) = (a \times b) \times c$

Hence, it is associative property.

3. The product of two rational numbers is always a _____

Sol. The product of two rational numbers is always a rational number.



NCF Intellectual Development
NEP Non - Perception

Multiple Choice Questions

true regarding the closure property of rational number
rational number.